

Department of Computer Science and Engineering

Islamic University of Technology (IUT)

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Laboratory Report

CSE 4412: Data Communication and Networking Lab

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Section: B

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Title: Configuring and Verifying of RIP and OSPF in a network topology.

Objective:

1. Describe the concept of dynamic routing.

2. Explain disadvantages of RIPv1 and improvement in RIPv2.

3. Configure Routing Information Protocol (RIP) in a network topology following given specifications.

4. Describe the concept of OSPF and related terminologies.

5. Explain the advantages of OSPF over RIP.

6. Configure OSPF in a network topology following the given specifications.

Devices/ software Used:

1. Laptop
2. Cisco Packet Tracer

Theory:

(Explain in brief the listed keywords)

1. Routing Information Protocol (RIP)

Ans: RIP is one of the oldest dynamic routing protocols used in computer networks. It's designed to allow routers to exchange routing information periodically within a network. RIP uses hop count as its metric to determine the best route to a destination network. Despite being obsolete in many modern network environments due to its limitations, RIP is still taught and understood because it provides a foundational understanding of dynamic routing protocols.

2. Forwarding Table used in RIP

Ans: In RIP, routers maintain a routing table, also known as a forwarding table, which contains information about reachable destinations in the network. This table includes entries for various networks along with associated metrics (typically hop counts in RIP) that indicate the cost of reaching those destinations. The router consults this table to determine the next hop for forwarding packets towards their destination.

3. Hop Count as cost
Ans: In RIP, the metric used to evaluate the "cost" of reaching a destination network is the hop count. Hop count refers to the number of routers (or hops) that a packet must traverse to reach the destination network. Each hop between routers typically incurs a certain amount of delay and resource consumption, so minimizing hop count generally leads to more efficient routing paths.

4. Timers in RIP
Ans: RIP employs several timers to manage the exchange of routing information and to detect network topology changes. The key timers in RIP include:

• Update Timer: This timer determines how frequently routers send routing table updates to their neighbors. In RIP, updates are sent every 30 seconds by default.

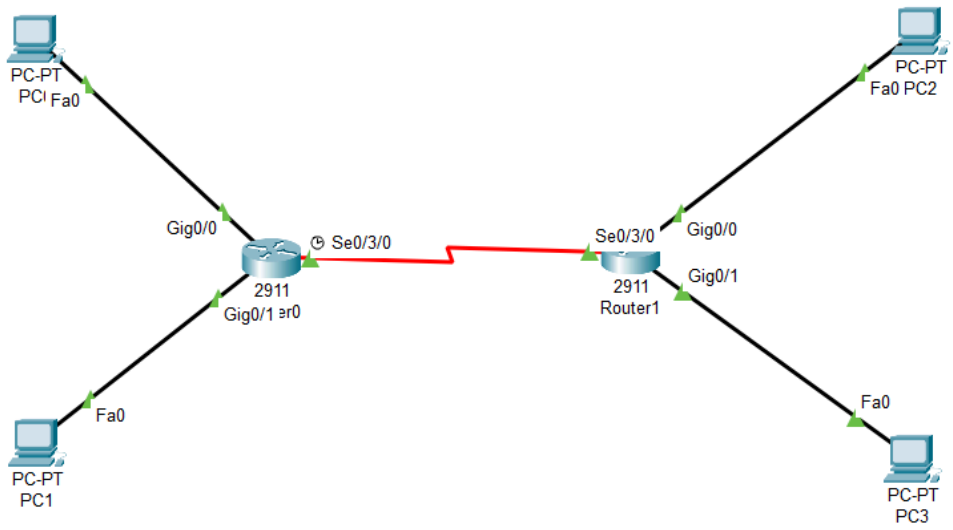
• Invalid Timer: When a router stops receiving updates about a particular route, it starts the invalid timer. If no updates are received for a route within a certain period (typically three times the update timer), the route is considered invalid.

• Hold-down Timer: After marking a route as invalid, the router starts the hold-down timer. During this time, the router suppresses any route updates received for the invalid route, preventing potentially incorrect information from being propagated.

• Flush Timer: Once the hold-down timer expires, indicating that the route is likely stable again, the router starts the flush timer. After the flush timer expires, the route is removed from the routing table if no updates confirming the route's validity are received.

Diagram of the experiment:

(Provide screenshot of the final network topology. Make sure to label the network components.)
Task #01:



Working Procedure:

(Explain in brief how you completed the tasks. Provide necessary screenshots of used commands for each task.)

Task #01:

Step 1: Configuring Routers

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 172.132.0.1 255.255.0.0
Router(config-if)#ip address 172.132.0.1 255.255.255.224
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#int gig0/0
Router(config-if)#ip address 172.134.0.1 255.255.255.128
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int gig 0/1
Router(config-if)#ip address 172.134.0.150 255.255.255.128
Router(config-if)#no shut

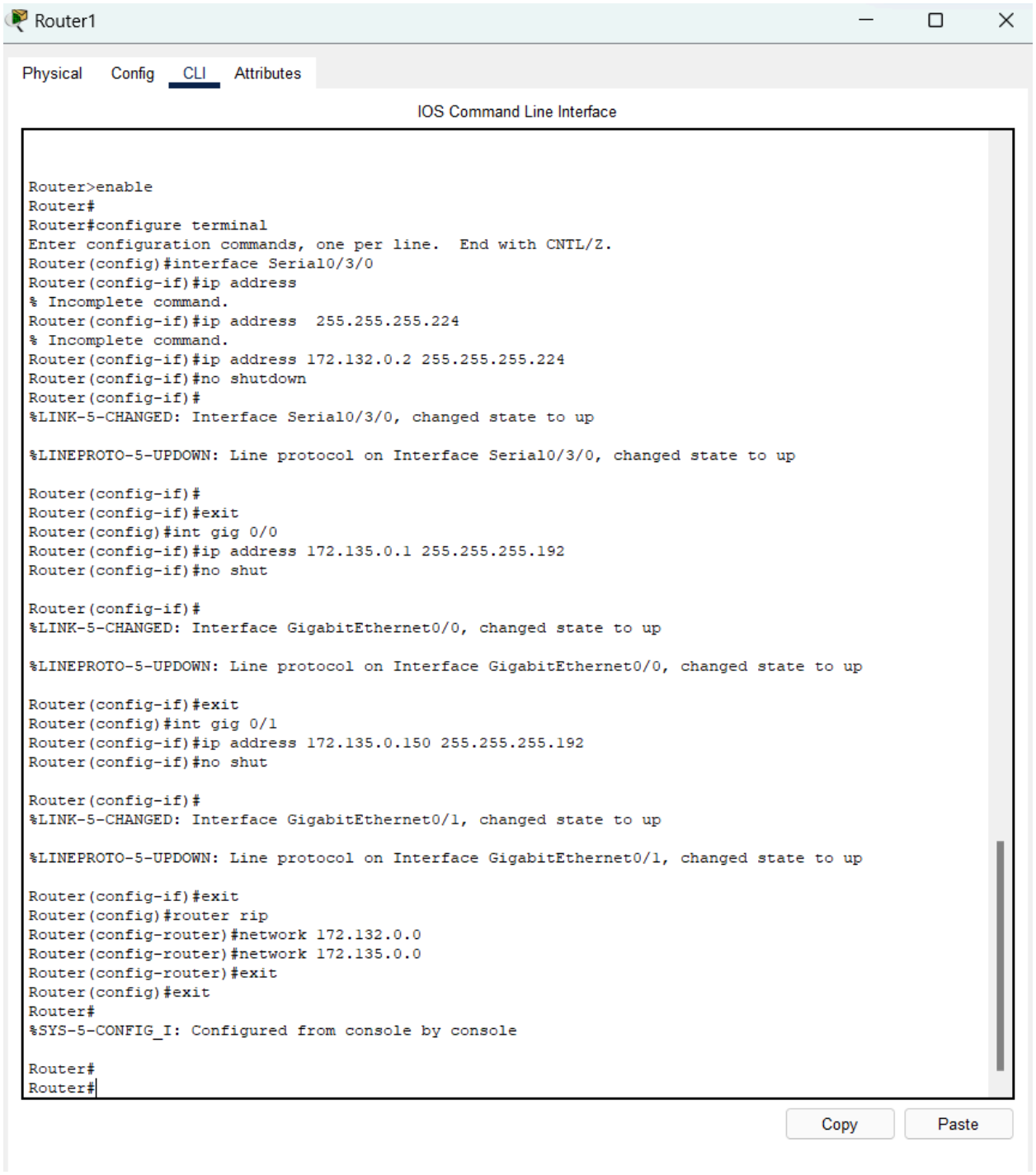
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

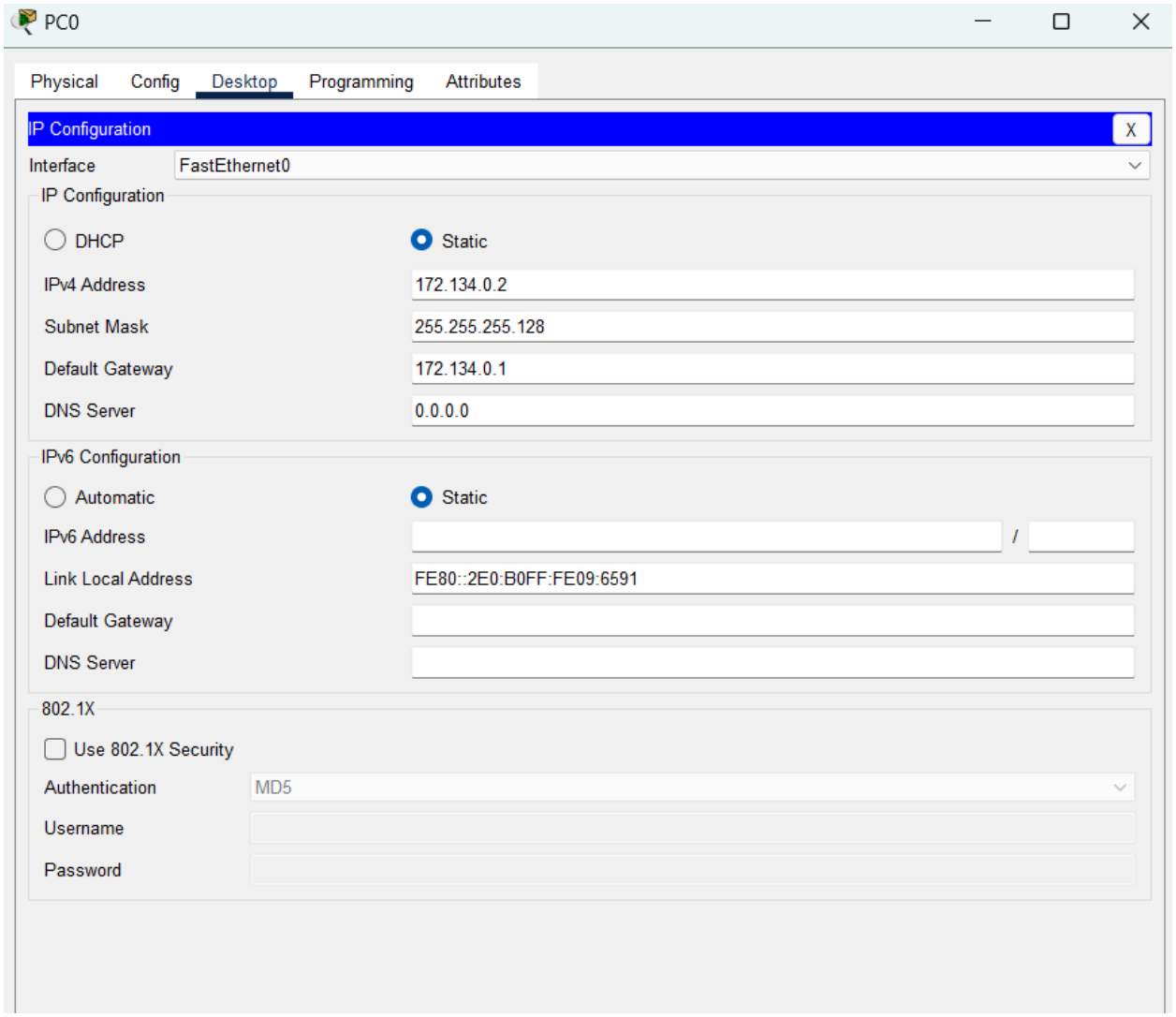
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#network 172.132.0.0
Router(config-router)#network 172.134.0.0
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

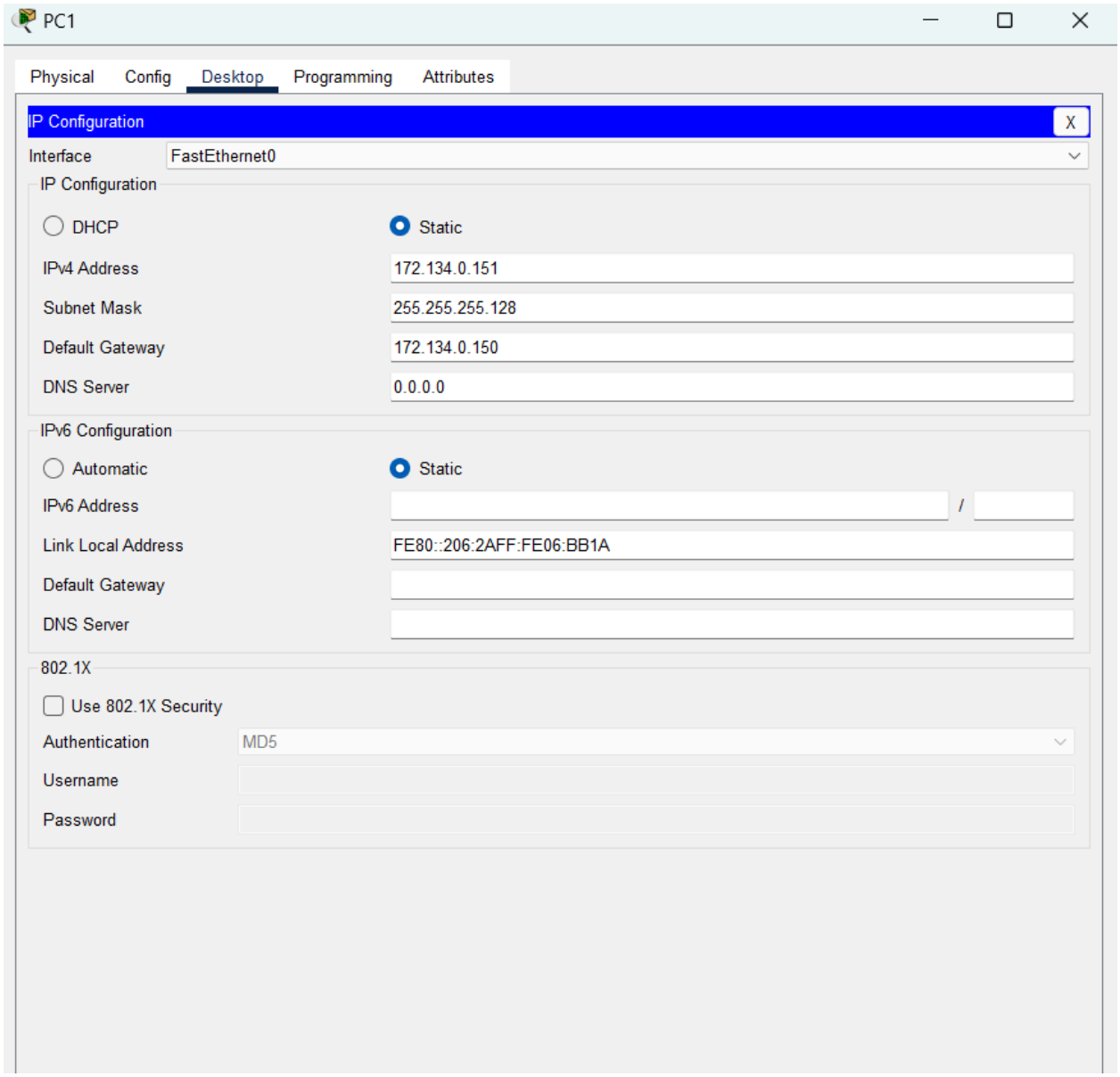
Copy Paste

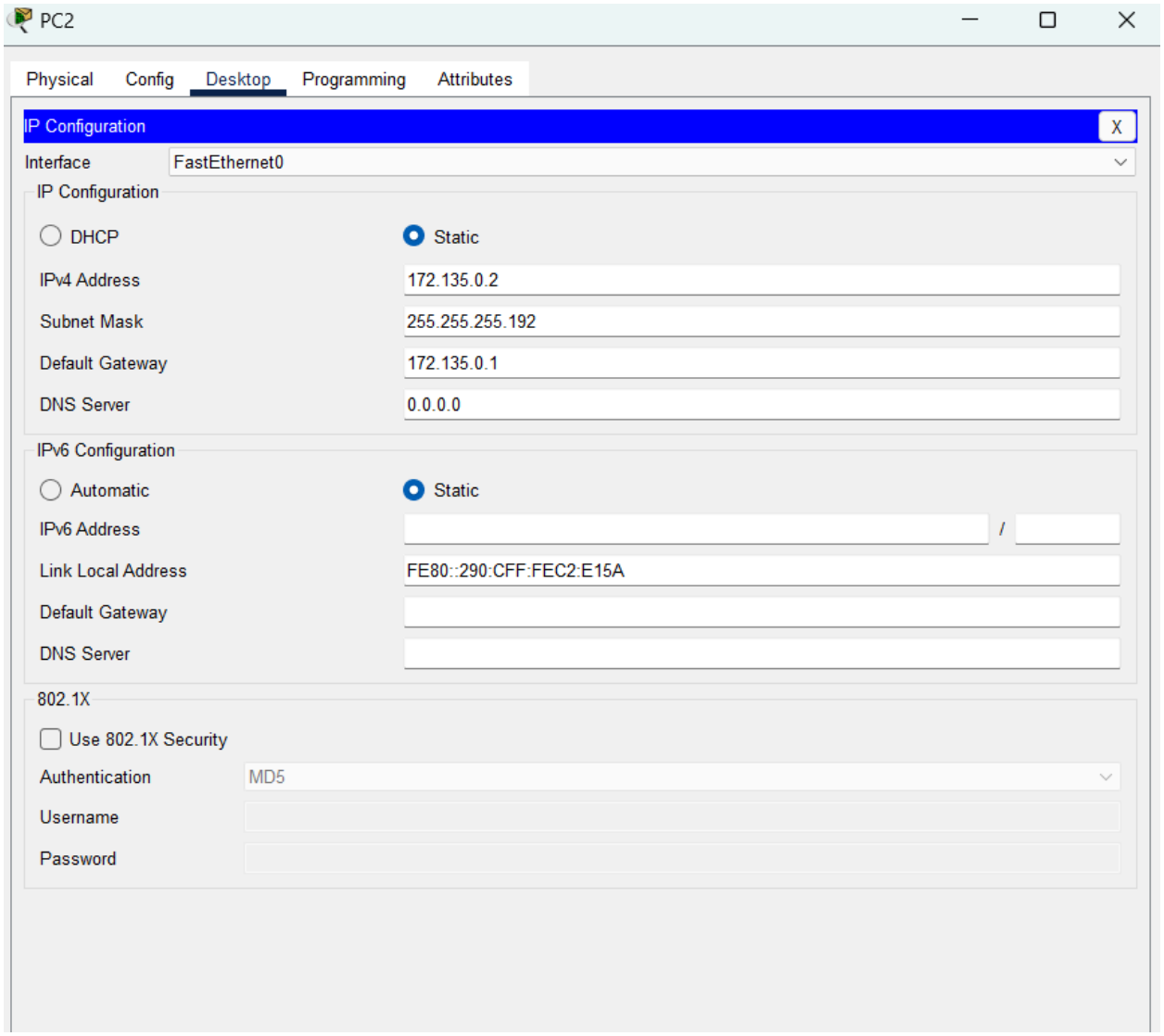
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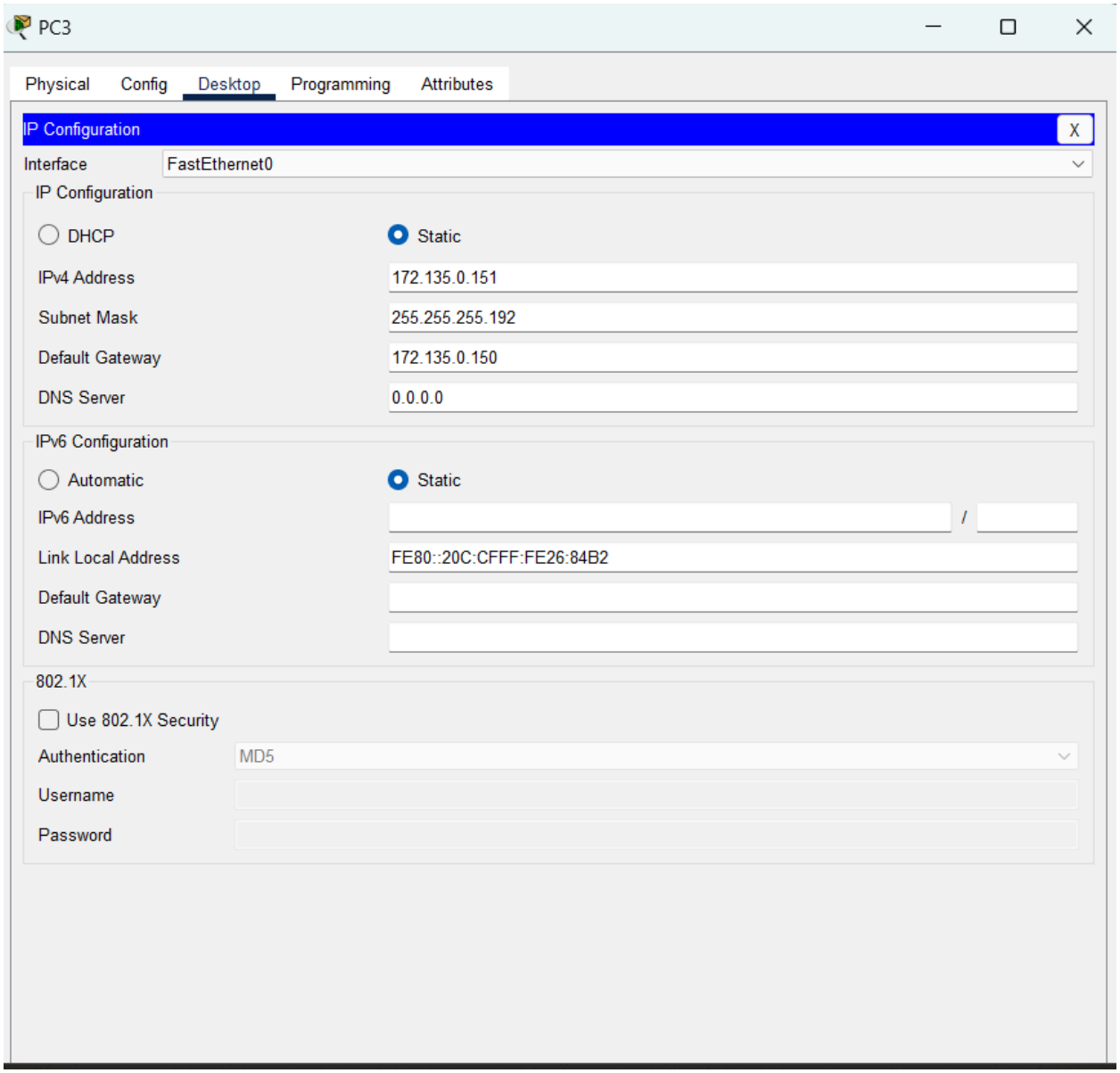


Step 2: Setting Up IPs


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.134.0.2
Subnet Mask
255.255.255.128
Default Gateway
172.134.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2E0:B0FF:FE09:6591
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.134.0.151
Subnet Mask
255.255.255.128
Default Gateway
172.134.0.150
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::206:2AFF:FE06:BB1A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.135.0.2
Subnet Mask
255.255.255.192
Default Gateway
172.135.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::290:CFF:FEC2:E15A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.135.0.151
Subnet Mask
255.255.255.192
Default Gateway
172.135.0.150
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::20C:CFFF:FE26:84B2
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password